

Wearable Technology Design and Accessibility Considerations Course

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Figure 1: Where on the body should wearable technology go to be useful and accessible?

ABSTRACT

The course proposed here will focus on design and accessibility considerations for wearable technology. In this course we will explore how to develop a robust set of design and accessibility considerations (guidelines) for wearable technology. The course will begin a presentation on wearability and accessibility then participants will engage in an activity using a new Wearable Technology Designer's Web Tool. The tool can be accessed again after the course and shared with students and colleagues. The course will end with a discussion about what design considerations might need to be

added to the tool and what human factors information or other research should be updated.

CCS CONCEPTS

• **Human-centered computing** → *Accessibility design and evaluation methods; Human computer interaction (HCI).*

KEYWORDS

Design, Accessibility, Web Tool

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1 INTRODUCTION

Whenever designers start out to create a piece of wearable technology there are some standard human factor considerations that we

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all have to think through. These might include where on the body it is the easiest to sense certain bio signals or types of movement, or they could include factors surrounding the social acceptability of interacting with on-body interfaces [1, 6]. It might be important to understand the human body's ability to feel and recognize vibration and on-body location [5, 9]. In any case academic literature on these human factors topics is broad and dispersed through the many different disciplines associated with wearable technology. Although there have been attempts to collect this information in survey papers on the subject of designing wearable technology [3, 7], new information come out quickly and compiling all the information necessary is difficult in the format of an academic conference article. These survey articles directed at the computing science (or other specific) community might also not have impact with the design community at large.

What is also important to consider at the beginning of a wearable technology project is how the decisions made about the interface and on-body location effect accessibility [8]. Poor decisions made in the beginning of a project can greatly effect accessibility and the ability of a large part of the population to use a wearable device. These decisions about interfaces and displays can be very hard to overcome at the end of a project and often can lead to additional costs and applique style add on accessibility solutions that do not seem cohesive with the original design.

Design and accessibility considerations presented in a simple way as guidelines pertaining to specific human factor concerns supported by academic literature could be very helpful to the design community.

2 WEARABLE TECHNOLOGY DESIGNER'S WEB TOOL

The *Wearable Technology Designer's Web Tool* is a tool that leads designers / developers through a series of questions about their wearable technology project. The questions are designed to force designers to think of all options at the beginning of a project, illuminating opportunities and shortcomings in accessibility caused by decisions along the way. As the designer makes selections pertaining to the purpose and interaction styles of their project a body map builds in real time showing where on the body the technology might best be placed. When all the questions are answered the tool creates a .pdf with the body map and all pertinent design and accessibility considerations for the project (including an extensive list of academic references for deeper research).

The *Wearable Technology Designers Web Tool* will be free to use. This beta version, and future versions, will be available after the workshop for all participants (and anyone else) to use in education and industry settings.

3 COURSE

This course will begin with a presentation on design and accessibility issues surrounding wearable technology and why they are important to consider at the beginning of a project. We will present case studies and prompt discussion from the participants about their own experiences.



Figure 2: Screen shot of the Wearable Technology Designer's Web Tool. The body map is located on the left and highlights the best locations on the body derived from the questions answered by the designer on the right

Participants will be asked to come to the course with an initial concept or wearable technology idea that would like to start designing. We will organize the participants in groups of 5 to work towards designing one of the group members ideas (or a combination of their ideas). The group will use the WTDWT to strategically think through design decisions. When the groups are finished we will meet back up as a whole to present concepts.

As groups present concepts we will ask that they speak to how the WTDWT might have made them consider their design and accessibility decisions more strategically.

As members of the workshop will be members of the CHI community and will be somewhat familiar with the research included in the WTDWT we will also illicit feedback about the tool. How might it be made better so that it would have a greater impact with industry? Is it set up appropriately for use in educational settings? Should there be changes to the user interface or display? Perhaps most importantly is there new research that should be included in the findings and calculations made by the web tool to produce the body maps?

3.1 Impact and Benefits

We hope that this course will spark a discussion about accessibility within the wearable technology design process. With the introduction of the *Wearable Technology Designer's Web Tool* WTDWT we hope to make early stages of the design process easier for designers and developers who might not have access or knowledge of the wealth of academic resources available on the many facets of wearable technology. In doing so we also hope to highlight accessibility concerns and train designers to consider all possible users. For this reason, we hope that the participants in the work shop will help us in discussion and surveys find even better ways to emphasize accessibility and perhaps illuminate ways the tool could be better.

We hope findings from the workshop discussion and surveys will be compiled and published as a white paper. New references and information will be included in the WTDWT.

3.2 Course Format and Intended Audience

The tutorial will focus on using the new Wearable Technology Designer's Web tool. We hope to accommodate up to 40 people if the teleconference software being used for CHI has the ability to create small groups for discussion with ease. Anyone interested in wearable technology, or teaching courses to students on wearable tech is encouraged to attend, no special skills required.

3.3 Course Schedule

Overview and Presentation – 30 min

Wearable Technology Designer's Web Tool

WTDWT Activity – 60 min

Presentations of Outcomes and Discussion – 60min

Collection of proposed Design and Accessibility Considerations and changes to WTDWT.

Collection of new academic references.

4 COURSE CALL FOR PARTICIPATION

We would like to invite you to participate in the CHI Wearable Technology Design and Accessibility Course. In this course we will explore how to develop a robust set of design and accessibility considerations (guidelines) for wearable technology. We will try out a new Wearable Technology Designer's Web Tool and discuss what features or information it might be missing. At the core of our workshop is an effort to understand how to train designers and developers of wearable technology to think about all users and accessibility.

There is no required submission to participate in this course, however we ask that you come to the workshop with a basic wearable technology idea or product you would like to create. We will form groups and work through the early stages of the design process for some of these ideas using the WTDWT.

Participants will receive a .pdf document of information tailored to their projects. Participants will be able to use the WTDWT for future projects and allow their students to use the tool as well. We hope through discussion to collect new information and considerations that might make the tool more useful.

5 COURSE ORGANIZERS

Clint Zeagler - Senior Research Scientist

Georgia Institute of Technology

[Clint Zeagler Bio](#)

Zeagler's research focuses on wearability and on-body interactions. He is the project lead for the Wearable Technology Designer's Web Tool WTDWT and is looking forward to the opportunity to have a robust discussion around design and accessibility considerations. Zeagler has successfully ran many workshops and focus groups as part of his role at the Institute for People and Technology including a workshop at ISWC in the past "Wear and Tear: Constructing Wearable Technology for the Real World". Zeagler has taught undergraduate and graduate level classes for the College of Computing and College of Design at Georgia Tech since 2007.

Brian Jones – Principal Research Scientist

Georgia Institute of Technology

Jones has developed interactive applications for use in informal learning environments, on smartphones, and in the home. In 2008, Mr. Jones was named Director of the Aware Home Research Initiative (AHRI) [4], a group of Georgia Tech faculty and students researching the next generation of technologies and applications to support residents in their homes. In this capacity, Jones is working to build new research and industry partnerships as well as enable faculty and students to innovate new technologies for the home that will improve the lives of residents. Jones works on making everyday interactions more accessible for the aging and people with disabilities. Jones will be able to market our workshop to those in the assistive technology community. Jones runs assistive technology workshops and teaches seminar courses at Georgia Tech.

Maribeth Gandy – Principal Research Scientist

Georgia Institute of Technology

Gandy has a rich history in wearable technology, accessibility, and tool kits for designers [2]. Her feedback in discussions and especially around how to incorporate participants feedback on the tool during discussion will be most helpful. She teaches courses in Audio Design, Augmented Reality, and User Interface design at Georgia Tech.

Scott Robertson - Senior Research Scientist

Georgia Institute of Technology

Robertson is the Technical lead on the WTDWT. Robertson has authored and co-authored refereed journal papers on a number of computer science topics, including video games for stroke rehabilitation, 3D visualization and VR surgery simulation, VR systems for simulating artistic and cultural performances, video game systems for air traffic control simulation training and human factors testing, telemedicine and distance learning. His expertise in designing and prototyping interactive systems will aid in discussion and a trusted and experienced resource.

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